Logic strategy guide (version 1.0.0)

The ungraded lessons shown at the start of each guided-strategy backward branch, by branch ID. Each worked example is a different puzzle from every scored question.

S1-B1 · Reflect one cell at a time

Strategy: reflect one cell at a time

A mirror line acts like a fold. Each point keeps its distance from the line but ends up on the other side. Points on the line stay where they are.

For a vertical mirror line, a cell keeps its row and only its column changes. For a horizontal mirror line, a cell keeps its column and only its row changes.

So reflect one cell at a time: count how far it is from the line, then count the same distance on the other side. Check two or three cells; if any one is wrong, the figure is not the reflection.

Worked example (a different figure)

A grid has 6 rows. A horizontal mirror line lies between rows 3 and 4. A shape has cells at row 1 column 2, row 2 column 2 and row 3 column 5. Where is its reflection?

1. Row 3 is 1 row above the line, so its image is 1 row below it: row 4. Row 2 goes to row 5 and row 1 to row 6.
2. Columns do not change in a horizontal mirror.
3. Image: row 6 column 2, row 5 column 2 and row 4 column 5.
4. Quick check: each cell and its image add up to 7 rows (1 + 6, 2 + 5, 3 + 4), the same for every pair, because the line is halfway between them.